



OLIST E-COMMERCE DATA WAREHOUSE

Data Modeling, Microsoft Fabric and Power BI

OLIST E-COMMERCE — DATA DOCUMENTATION

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1. Data Overview

The Dataset

Market

Olist is an online e-commerce store which sells and delivers items all over Brazil

Period

Orders placed between September 2016 and October 2018

Volume

~99K orders ~112K items. ~99K reviews

Source file

8 CSV files

Source file	Rows
olist_orders_dataset	99,441
olist_customers_dataset	99,441
olist_order_items_dataset	112,650
olist_order_payments_dataset	103,886
olist_order_reviews_dataset	99,224
olist_products_dataset	32,951
olist_sellers_dataset	3,095
product_category_name_translation	71

every table column-by-column

olist_orders_dataset

Column	Description
order_id	Unique order identifier
customer_id	Order-scoped customer alias -> customers.customer_id
order_status	Lifecycle: created, approved, invoiced, processing, shipped, delivered, canceled, unavailable
order_purchase_timestamp	When the customer placed the order
order_approved_at	Payment approval moment
order_delivered_carrier_date	Handover to the logistics carrier
order_delivered_customer_date	Actual delivery to the customer
order_estimated_delivery_date	Promised delivery estimate shown at checkout

olist_customers_dataset

Column	Description
customer_id	Per-order customer alias (1:1 with orders)
customer_unique_id	The real person identifier (96,096 unique people)
customer_zip_code_prefix	First 5 digits of Brazilian CEP
customer_city	City at purchase time
customer_state	State UF code (27 values)

olist_order_items_dataset

Column	Description
order_id	Order this line belongs to
order_item_id	Line sequence inside the order (1-21)
product_id	Product sold
seller_id	Seller fulfilling this line
shipping_limit_date	Seller SLA deadline to hand to carrier
price	Item price at point of sale (BRL)
freight_value	Freight charged for this line (BRL)

olist_order_payments_dataset

Column	Description
order_id	Order being paid
payment_sequential	Payment line sequence
payment_type	credit_card, boleto, voucher, debit_card, not_defined
payment_installments	Installment count (1-24)
payment_value	Amount of this payment line (BRL)

olist_order_reviews_dataset

Column	Description
review_id	Review identifier — NOT unique (814 duplicates!)
order_id	Reviewed order
review_score	1-5 star rating
review_comment_title	Optional comment title

review_comment_message	Optional free-text comment
review_creation_date	When the review form was sent
review_answer_timestamp	When the customer answered

olist_products_dataset

Column	Description
product_id	Unique product identifier
product_category_name	Category in Portuguese
product_name_lenght	Chars in product name
product_description_lenght	Chars in description
product_photos_qty	Number of published photos
product_weight_g	Weight in grams
product_length_cm / height_cm / width_cm	Package dimensions

olist_sellers_dataset

Column	Description
seller_id	Unique seller identifier
seller_zip_code_prefix	Seller CEP prefix
seller_city	Seller city
seller_state	Seller State

product_category_name_translation

Column	Description
product_category_name	Category in Portuguese
product_category_name_english	Category in English

1.1 Analytical Questions

The dashboard is not a set of charts — it is an answering machine. Each visual exists to answer one of the questions below.

Q1. What is our revenue growth performance ?

Q2. Which customer states generated the highest total revenue and order volume?

Q3. How does delivery performance vary across states?

Q4. What is the overall On-Time Delivery % and average delay duration for late orders?

Q5. Which specific states suffer from severe delivery delays to optimize logistics partnerships?

Q6. What is the current Average Review Score and the overall percentage of negative reviews?

Q7. Which product categories achieve the highest (and lowest) customer review ratings?

Q8. How does the number of published product photos (`product_photos_qty`) correlate with overall sales volume ?

Q9. How efficiently are sellers performing, and which seller cohorts contribute the most to top-line revenue and service quality?

Q10. Who are the top-performing sellers that require strategic retention and incentives?

Q11. • Does high seller volume correlate with logistics speed and review scores?

Q12. Which product categories drive the most revenue and volume to prioritize inventory management and promotional campaigns?

Q13. Which high-freight SKUs need packaging/shipping optimization to cut costs?

Q14. What is the breakdown of payment types (`credit_card`, `boleto`, `voucher`) by total volume and value?

Q15. What is the Average Installments count chosen by customers on credit card transactions?

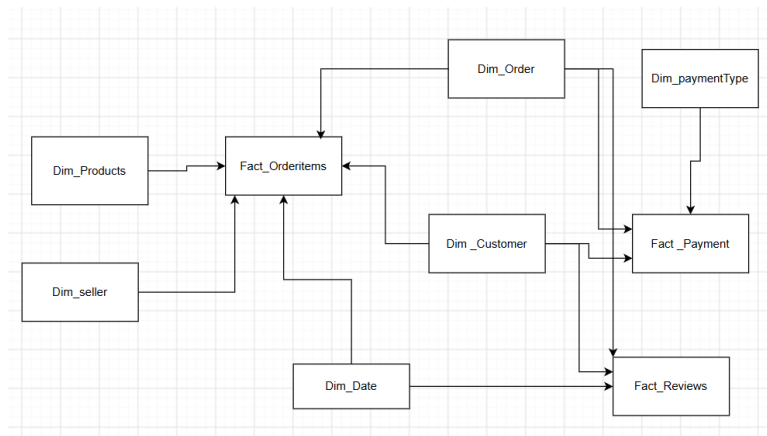
Q16. How do payment method preferences vary across different geographic states?

2. Conceptual, Logical & Physical Model

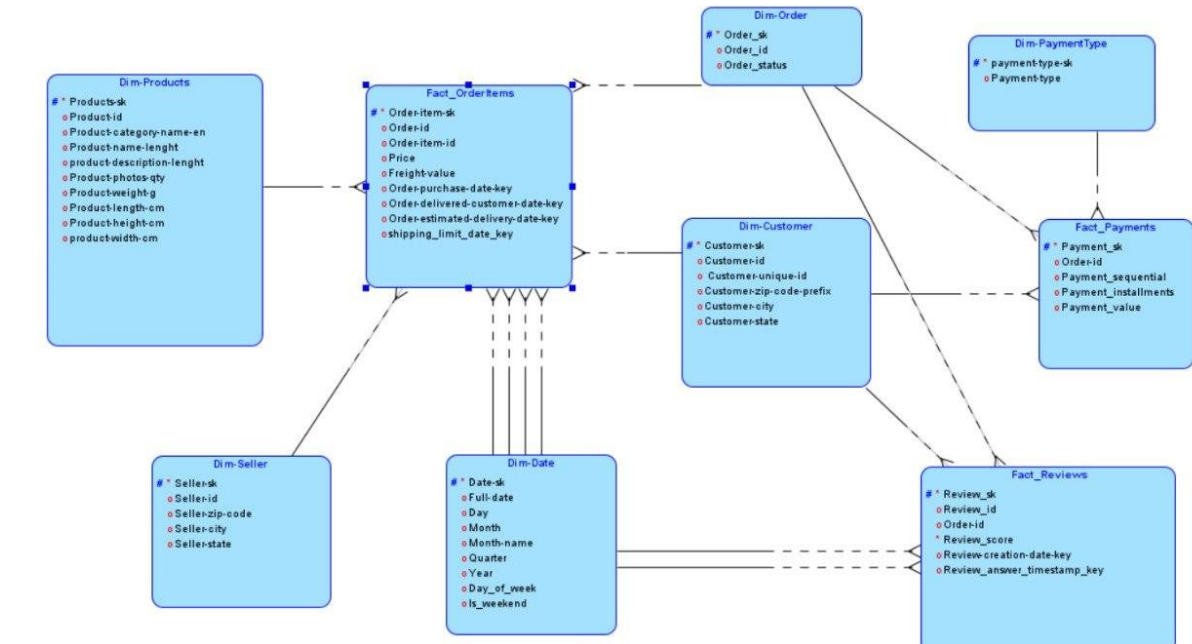
Schema Type: Fact Constellation Schema (Galaxy Schema)

The model follows a **Fact Constellation Schema** structure, featuring **three central fact tables** (Fact_OrderItems, Fact_Payments, and Fact_Reviews) that share multiple surrounding dimension tables (such as Dim-Products, Dim-Customer, Dim-Date, Dim-Seller, Dim-Order, and Dim-PaymentType) to provide context across different business processes.

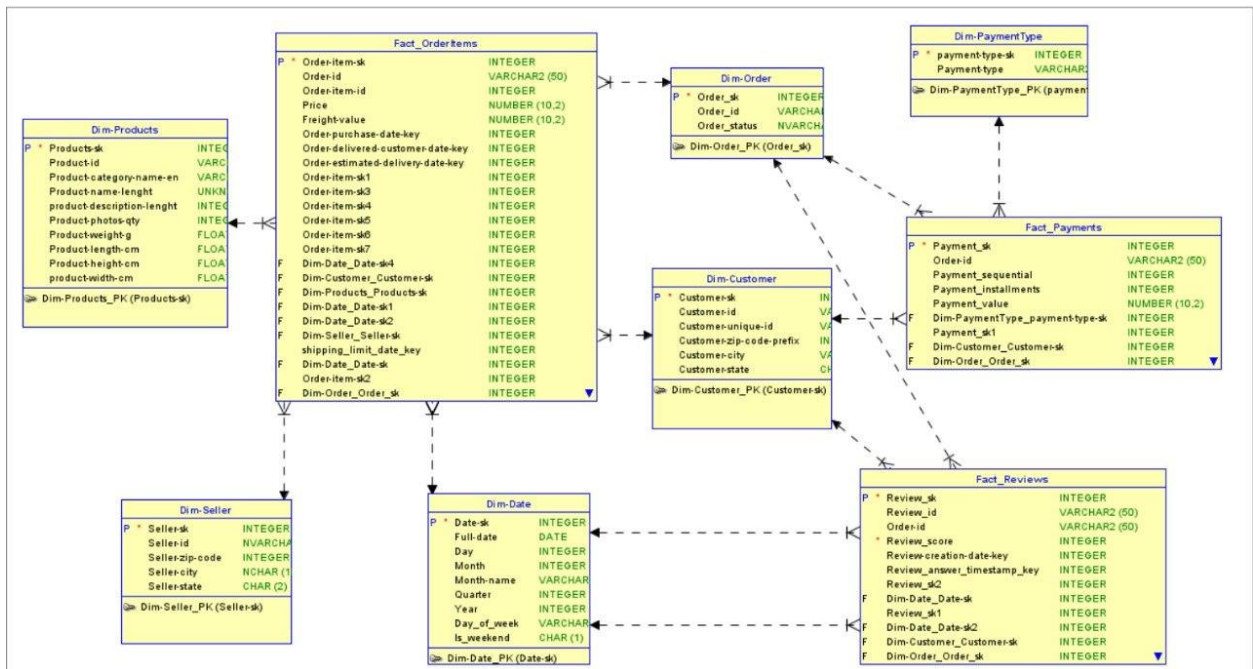
2.1 Conceptual Model



2.2 Logical Model



2.3 Physical Model



Olist E-Commerce Bus Matrix

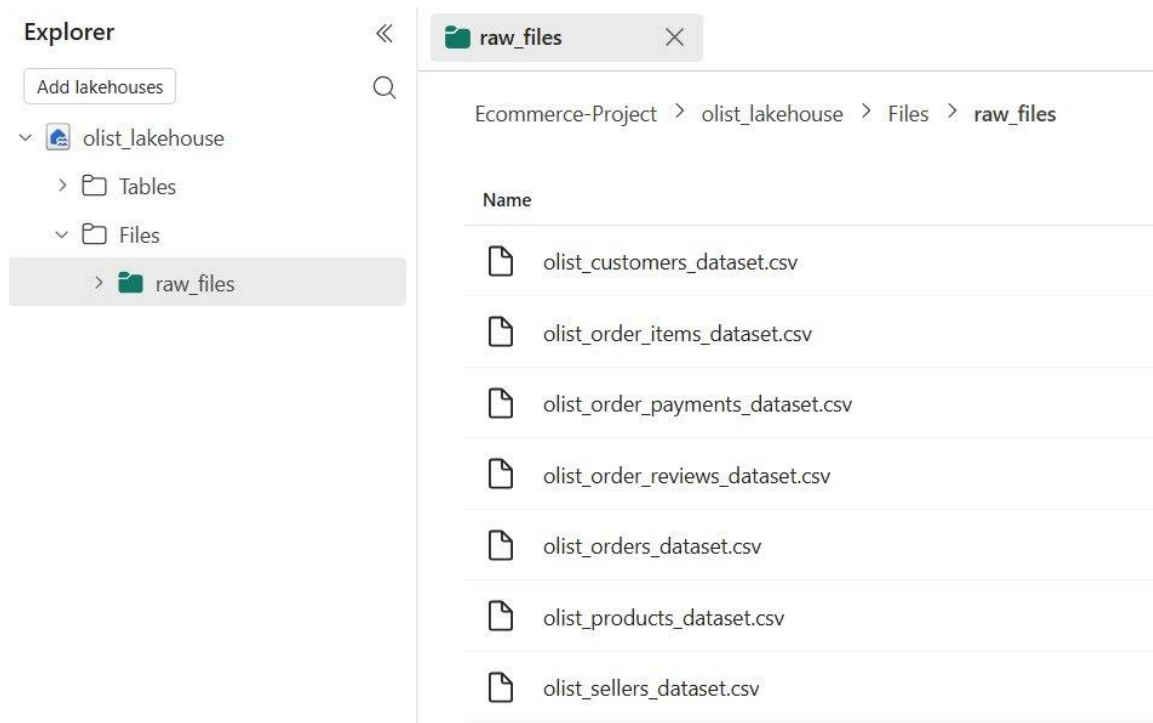
Dimension	Fact_OrderItems	Fact_Payments	Fact_Reviews
Dim-Products	✓	✗	✗
Dim-Seller	✓	✗	✗
Dim-Date	✓	✗	✓
Dim-Order	✓	✓	✓
Dim-Customer	✓	✓	✓
Dim-PaymentType	✗	✓	✗

3. Microsoft Fabric Pipeline & Data Integration (ETL)

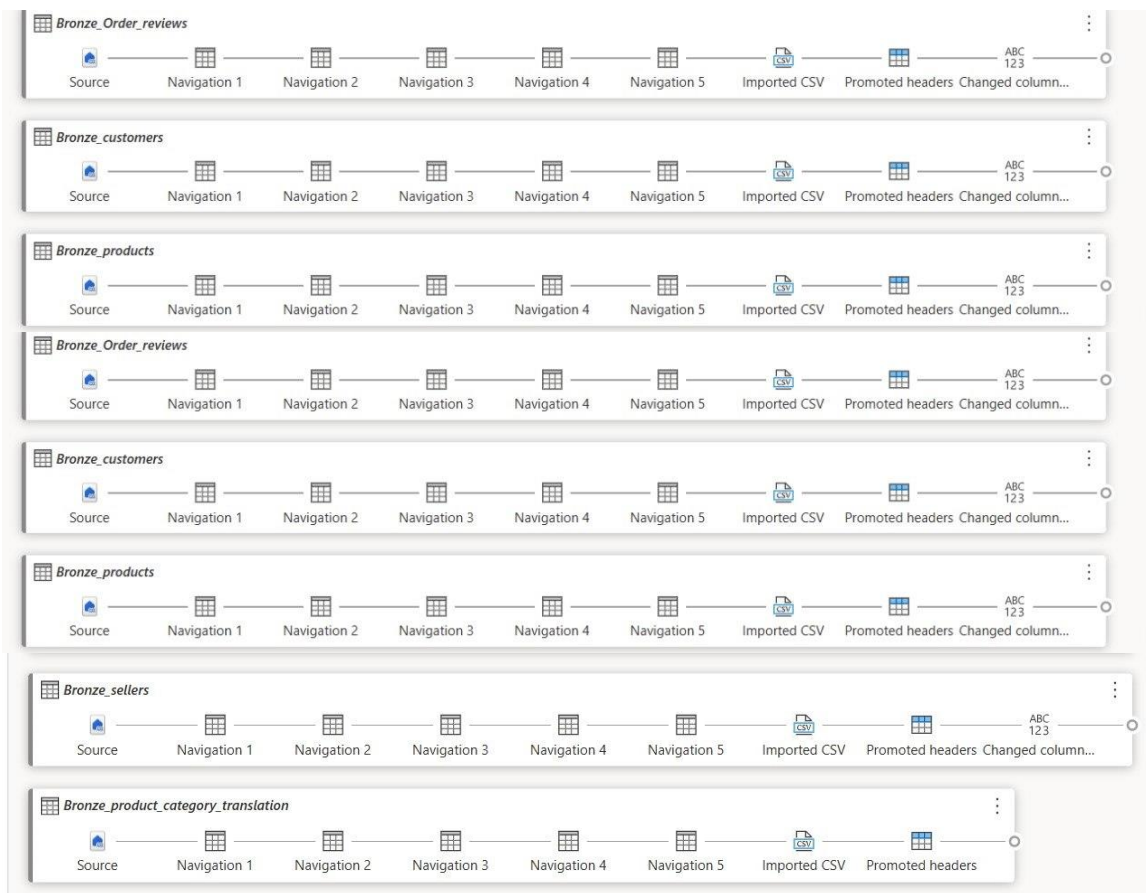
In this project, a Medallion Architecture (Bronze --->Silver----->Gold) was implemented using Microsoft Fabric Data Factory (Dataflows Gen2 and Pipelines) to extract, clean, transform, and load raw e-commerce data into a central Data Warehouse.

3.1 Bronze (raw)

- **Setup Data Lakehouse:** Created a Lakehouse named (olist_lakehouse) to serve as the landing storage zone for raw data.
- **Raw File Upload:** Uploaded 8 source CSV files (e.g., olist_orders_dataset.csv, olist_customers_dataset.csv, olist_order_items_dataset.csv, olist_order_payments_dataset.csv, olist_order_reviews_dataset.csv, olist_products_dataset.csv, olist_sellers_dataset.csv, and product_category_name_translation.csv) into the Lakehouse Files directory (raw_files).
- **Dataflow Gen2 (DF_Bronze_Ecommerce):** Created a Dataflow Gen2 to ingest raw files, set appropriate headers, enforce initial column data types, and load them as raw Delta tables (Bronze_*) inside the Lakehouse staging area.



product_category_name_translation.csv



3.2 Silver (Cleansing & Standardization)

Dataflow Gen2 (DF_Silver_Ecommerce): Built a Silver Dataflow Gen2 referencing the Bronze tables to perform data cleansing, normalization, and prep for dimensional modeling:

- **Text Trimming & Standardization:** Cleaned text fields across tables (e.g., trimming whitespace in customer cities, states, and order statuses).
- **Deduplication:** Identified and removed duplicate records (e.g., handling duplicate review entries in olist_order_reviews_dataset and customer IDs).

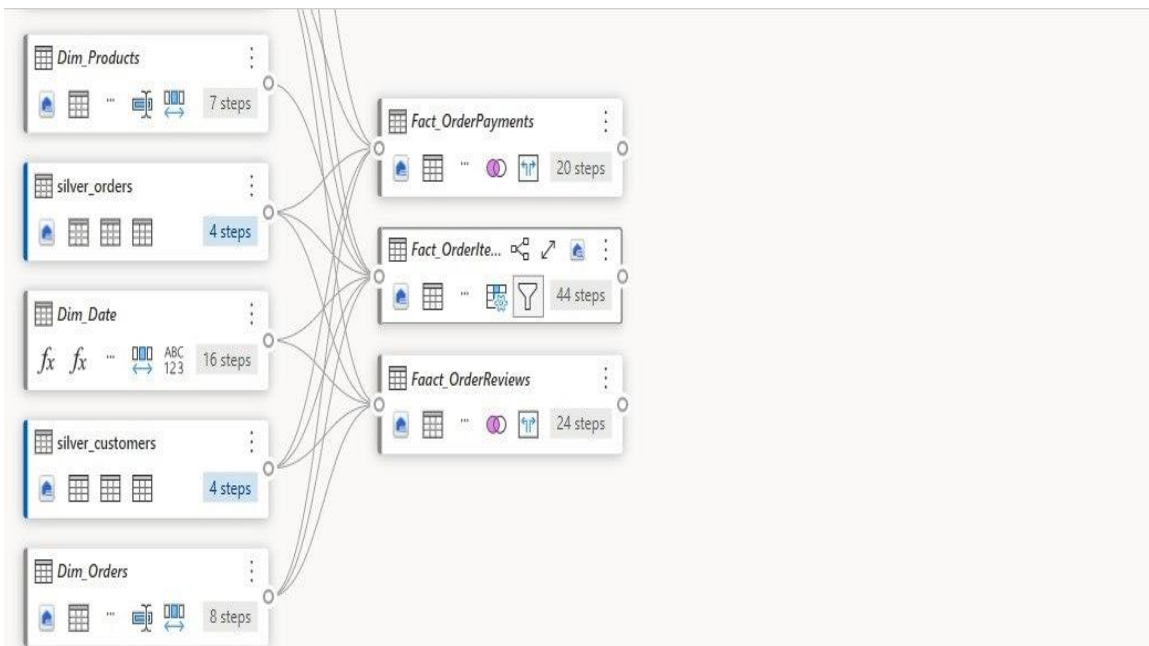
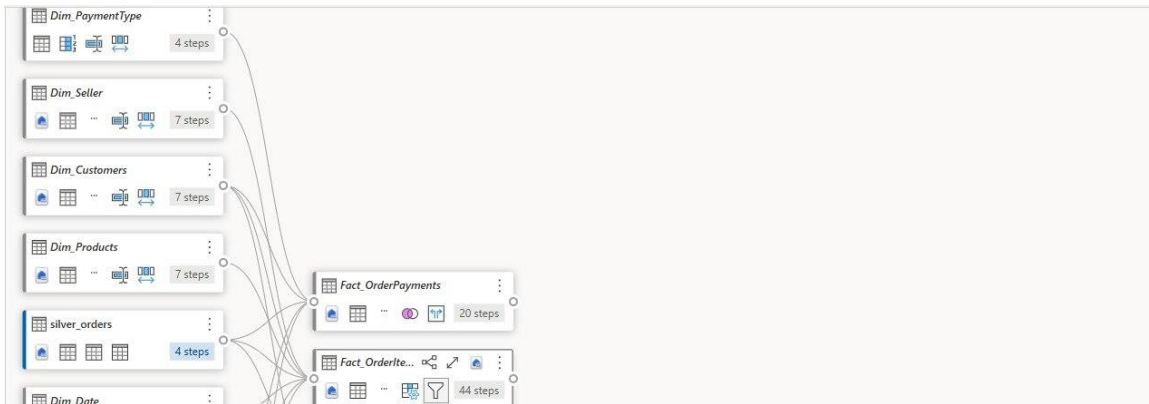
- **Category Name Translation:** Merged olist_products_dataset with product_category_name_translation to replace Portuguese category names with English equivalents (product_category_name_english).



3.3Gold (Dimensional Modeling& DataWarehouse Loading)

Dataflow Gen2 (DF_Gold_Ecommerce): Developed a Gold Dataflow Gen2 to transform cleansed Silver data into a **Fact Constellation / Galaxy Schema** structure within the Microsoft Fabric Data Warehouse:

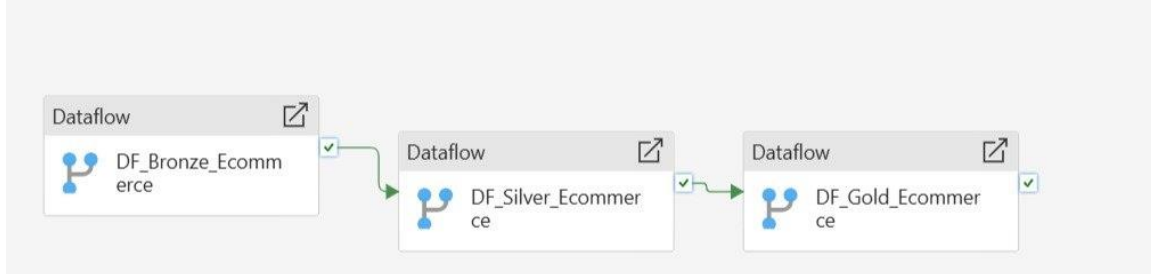
- **Dimension Creation:** Generated surrogate key IDs and built standard dimension tables: Dim_Products, Dim_Seller, Dim_Customers, Dim_Orders, Dim_PaymentType, and a generated Dim_Date calendar table.
- **Fact Tables Construction & Joins:** Built 3 central Fact tables (Fact_OrderItems, Fact_Payments, Fact_Reviews) by joining Silver transaction tables with the corresponding dimensions to map FK surrogate keys.
- **Data Warehouse Storage:** Loaded all 6 Dimension tables and 3 Fact tables into the Fabric Data Warehouse for downstream Semantic Modeling and analytical querying.



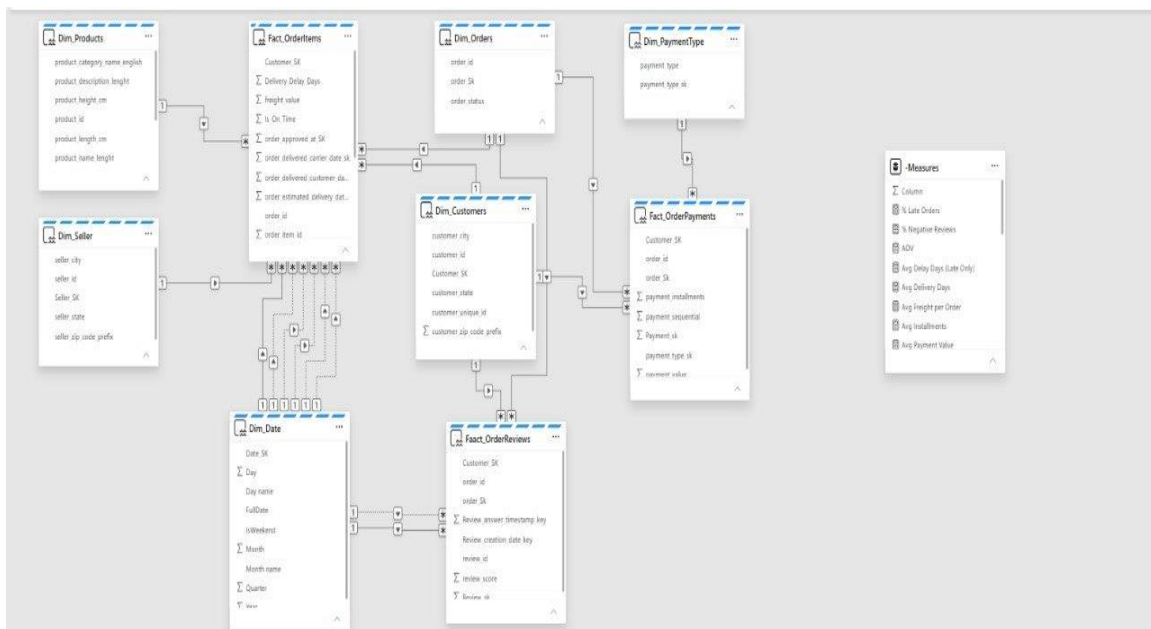
3.4Orchestration

Built an automated Fabric Data Pipeline to sequence and orchestrate the execution flow across the medallion architecture in order:

[DF_Bronze_Ecommerce] --→ [DF_Silver_Ecommerce] -> [DF_Gold_Ecommerce]



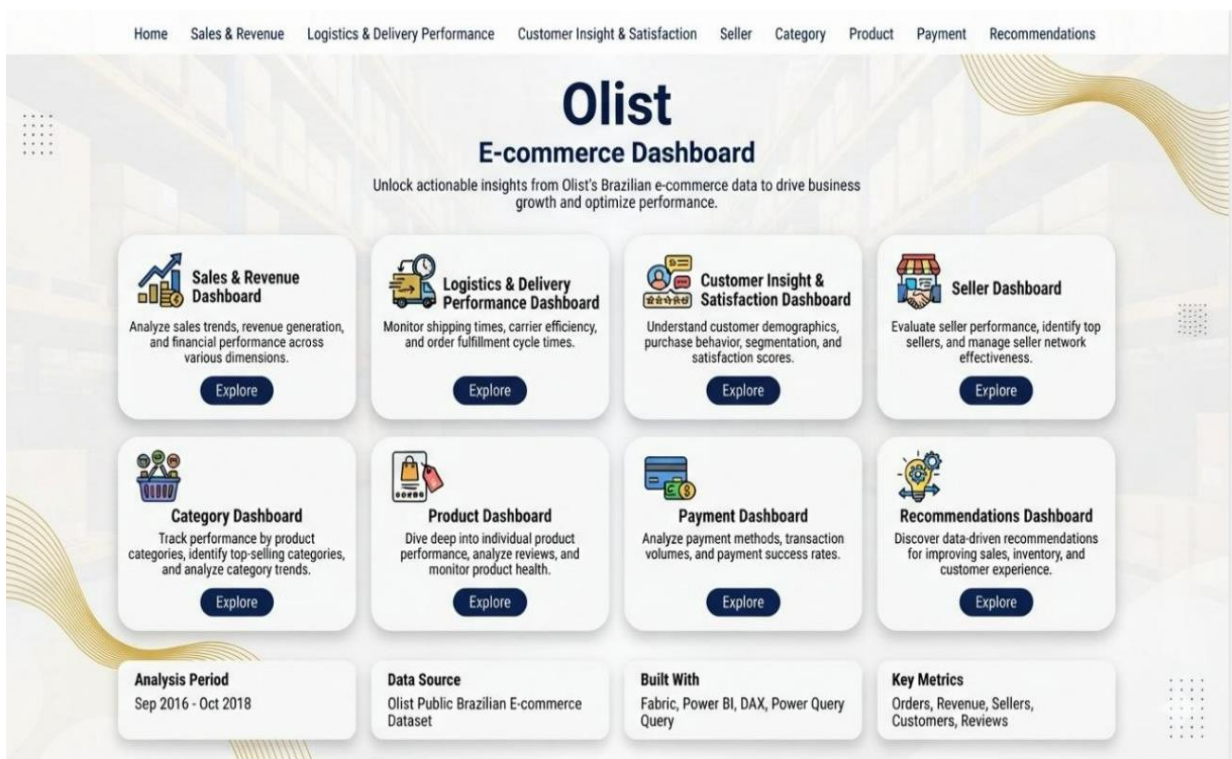
3.5Semantic Model

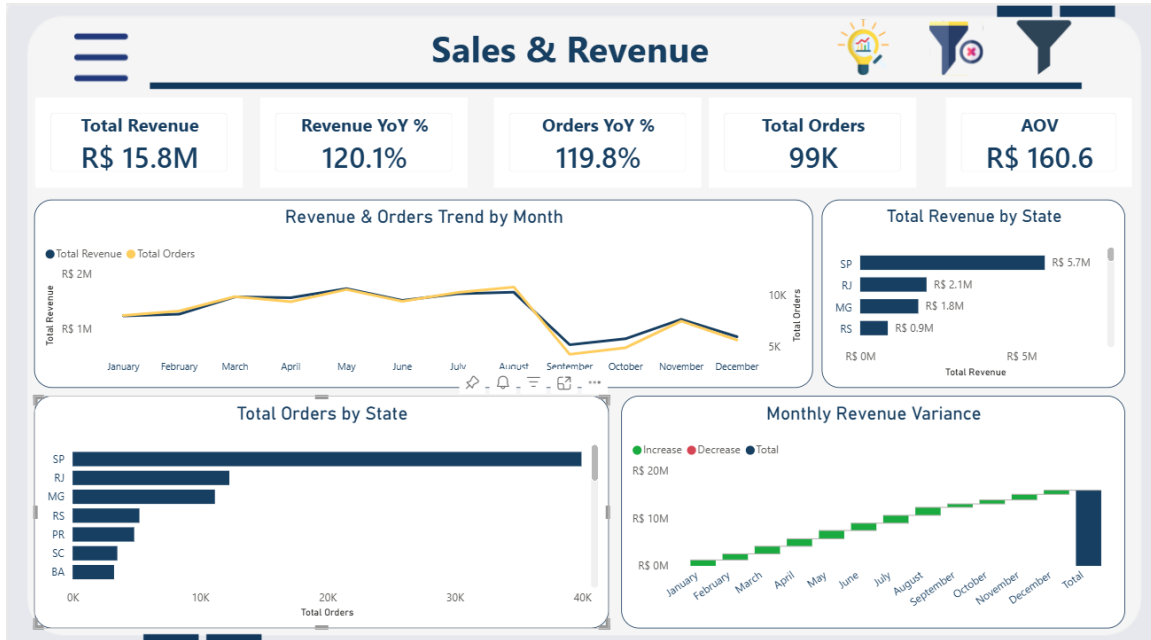


Core DAX measures :

Measure Name	Dax Expression
Total Revenue	Total Revenue = SUM(Fact_OrderPayments[payment_value])
Total Orders	Total Orders = DISTINCTCOUNT(Dim_Orders[order_id])
Total Customers	Total Customers = DISTINCTCOUNT(Dim_Customers[customer_unique_id])
Total Sellers	Total Sellers = DISTINCTCOUNT(Dim_Seller[seller_id])
AOV (Average Order Value)	AOV = DIVIDE([Total Revenue], [Total Orders], 0)
Avg Delivery Days	Avg Delivery Days = AVERAGE(Dim_Orders[Delivery_Days])
Avg Delay Days (Late Only)	Avg Delay Days (Late Only) = CALCULATE(AVERAGE(Dim_Orders[Delay_Days]), Dim_Orders[Delay_Days] > 0)
Avg Review Score	Avg Review Score = AVERAGE(Fact_OrderReviews[review_score])
Avg Revenue per Seller	Avg Revenue per Seller = DIVIDE([Total Revenue], [Total Sellers], 0)
% Late Orders	% Late Orders = DIVIDE(CALCULATE([Total Orders], Dim_Orders[Is_Late] = TRUE()), [Total Orders], 0)
On-Time Delivery %	On-Time Delivery % = 1 - [% Late Orders]
Repeat Customers	Repeat Customers = CALCULATE(COUNTROWS(FILTER(VALUES(Dim_Customers[customer_unique_id]), CALCULATE([Total Orders]) > 1)))
Repeat Purchase Rate	Repeat Purchase Rate = DIVIDE([Repeat Customers], [Total Customers], 0)
Seller YoY %	Seller YoY % = DIVIDE([Total Revenue] - [Seller Revenue LY], [Seller Revenue LY], 0)
% Negative Reviews	DIVIDE(CALCULATE(COUNTROWS(Faact_OrderReviews), Faact_OrderReviews[review_score] <= 2), COUNTROWS(Faact_OrderReviews))
Avg Installments	AVERAGE(Fact_OrderPayments[payment_installments])
Avg Product Price	DIVIDE(SUM(Fact_OrderItems[Price]), COUNTROWS(Fact_OrderItems), 0)

5. Power BI Overview





Year

All
▼

Month name

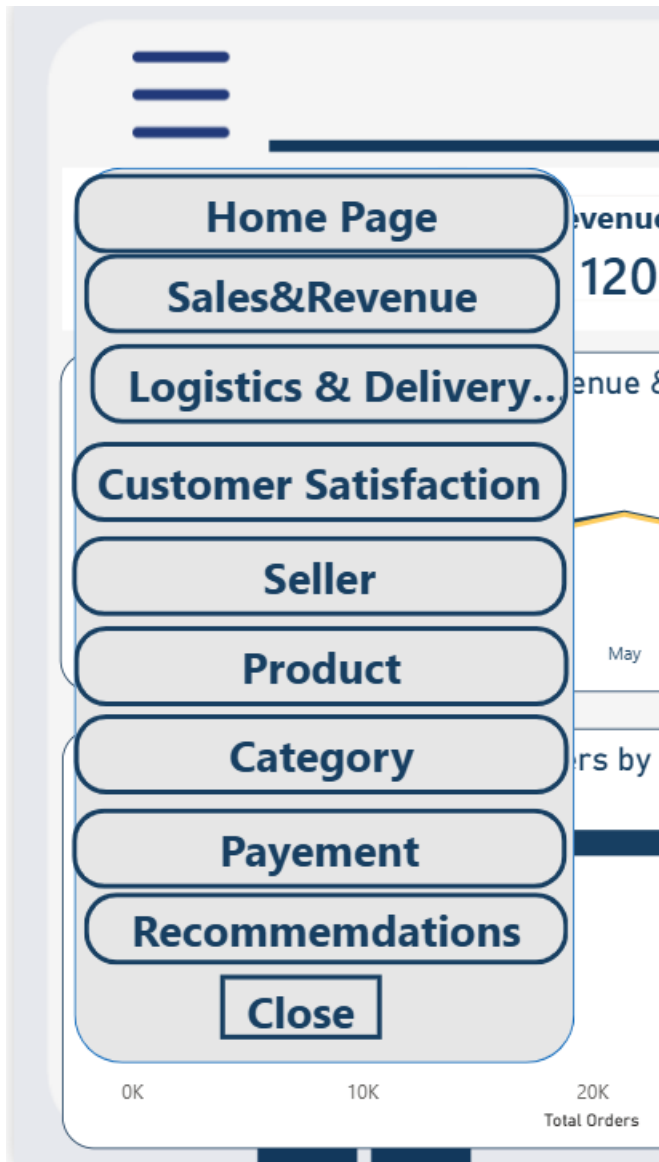
All
▼

Quarter

All
▼

customer_state

All
▼



Sales & Revenue



Executive Insights | Sales & Revenue

X

Overall Performance: Generated **R\$ 15.8M** in total revenue across **99K orders**, with an Average Order Value (AOV) of **R\$ 160.6**.

Annual Growth: Strong performance with Revenue YoY growth of **120.1%** and Order Volume YoY growth of **119.8%**.

Geographic Breakdown: São Paulo (SP) continues to lead as the top revenue contributor (**R\$ 5.7M**), followed by RJ (**R\$ 2.1M**) and MG (**R\$ 1.8M**).

rch

To

● Increase ● Decrease ● Total

State & City Details

Details by

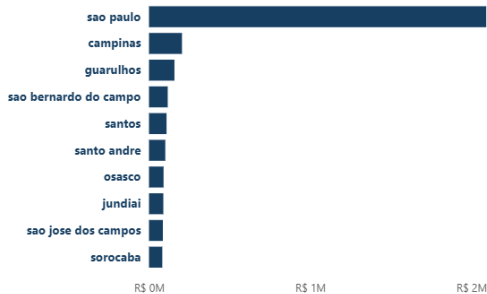
SP

Total Orders
40K

Avg Rev Score
4.2

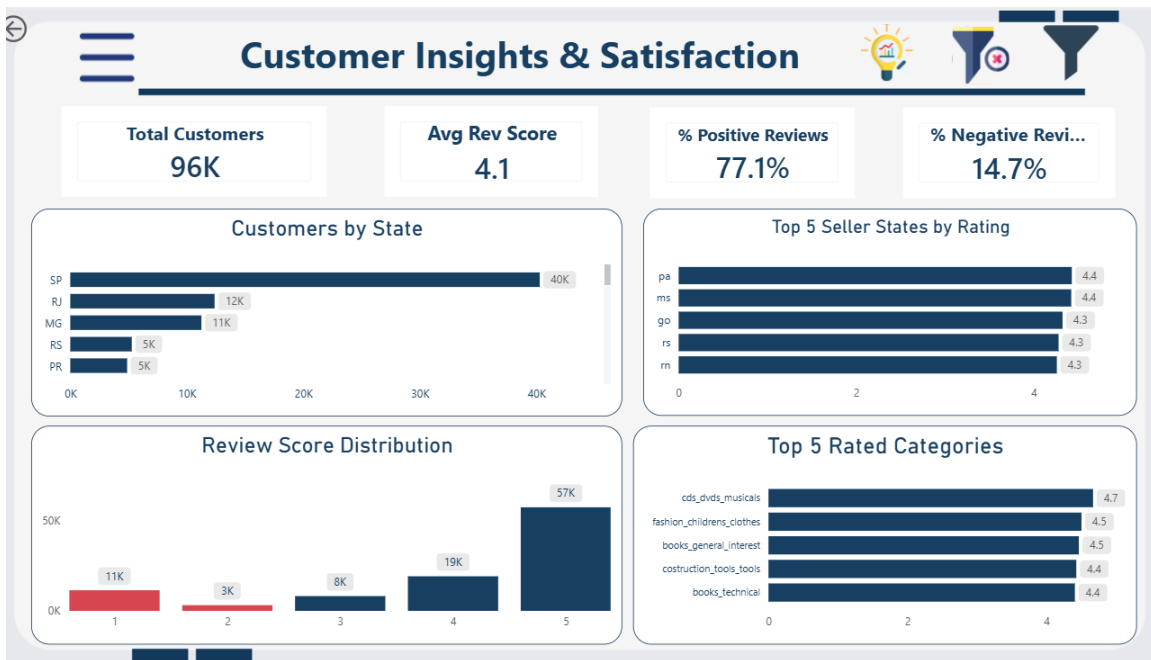
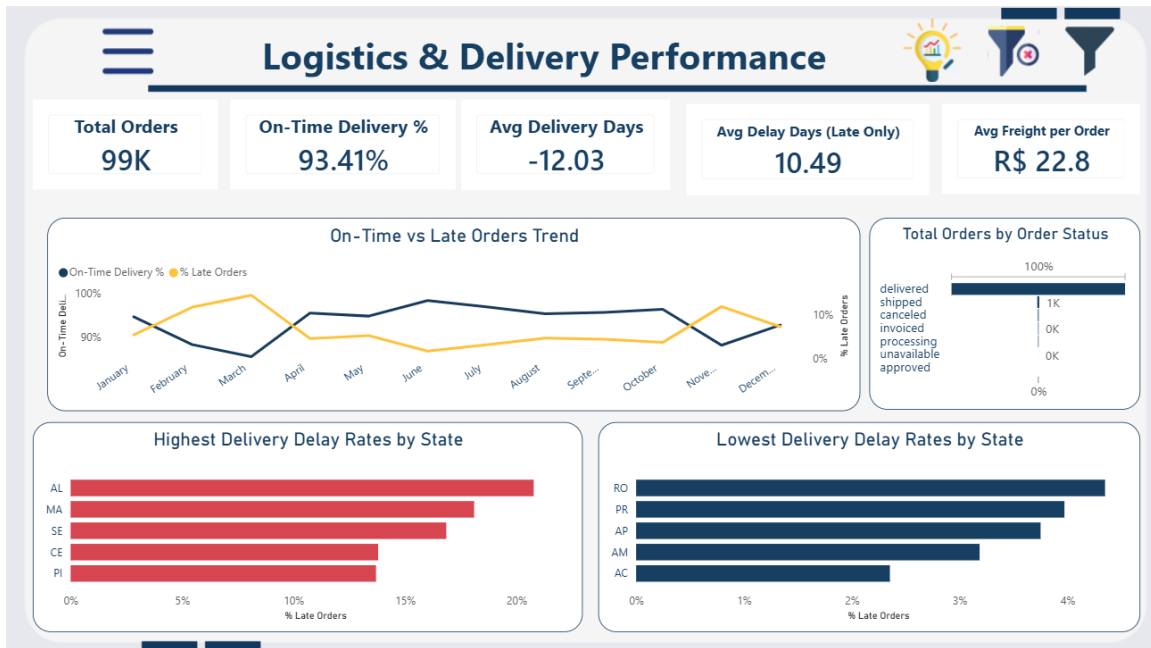
Avg Delivery Days
-11.16

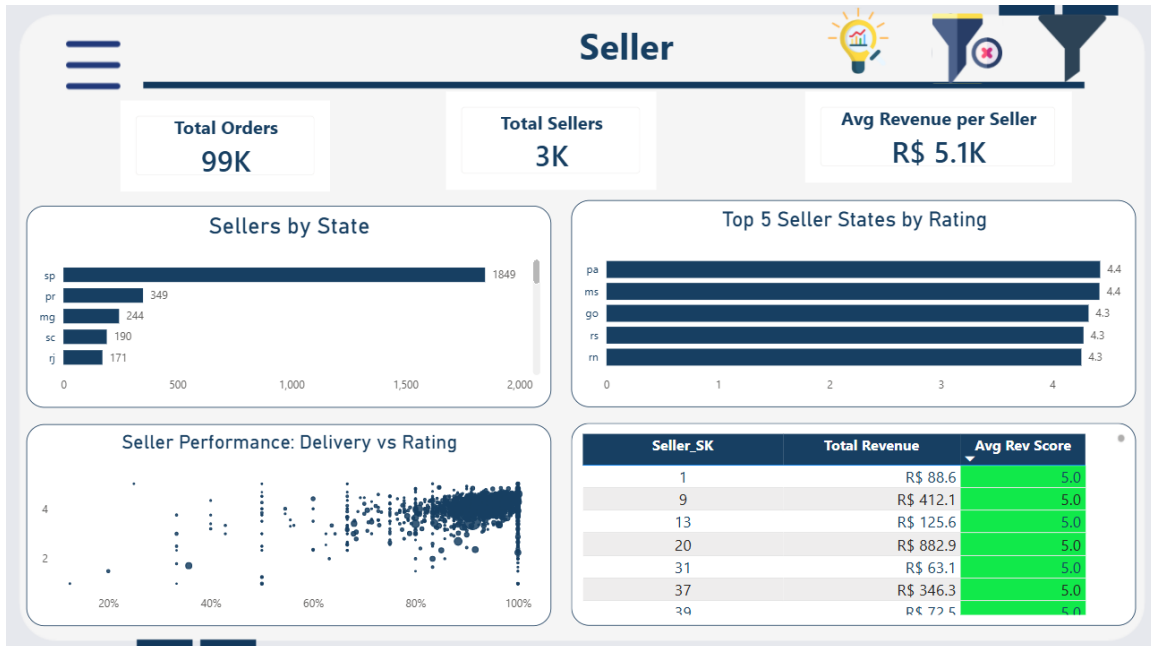
Top 10 Cities by Revenue

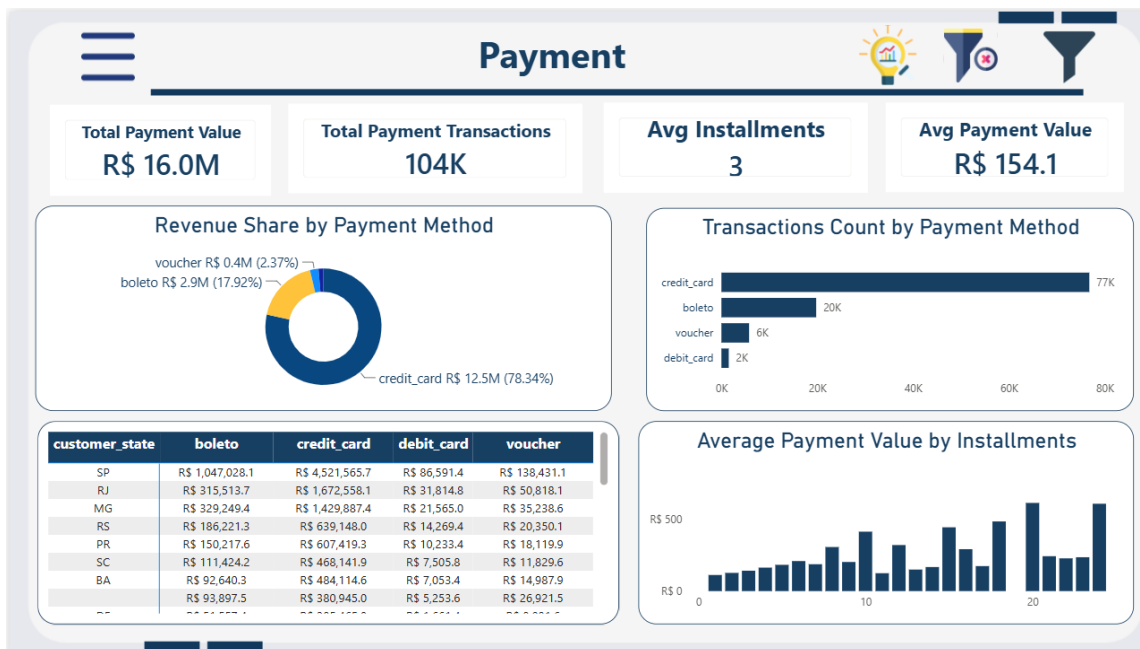
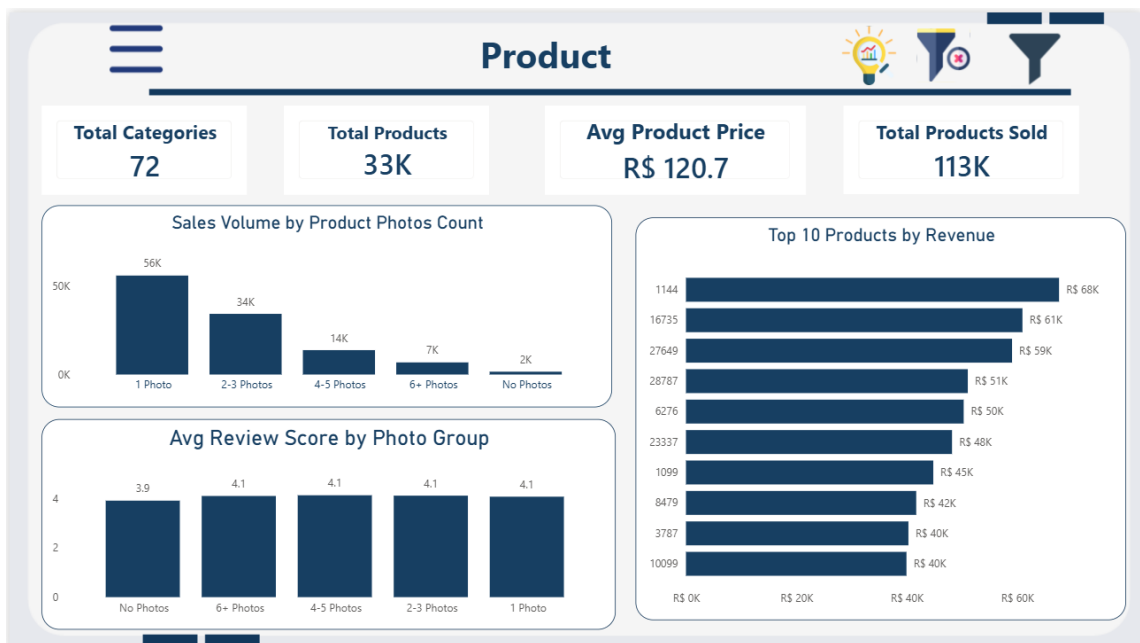


City Performance Details

customer_city	Total Revenue	Total Orders	Avg Delivery Days
sao paulo	R\$ 2,092,366.4	14830	-10.74
campinas	R\$ 204,637.3	1384	-9.75
guarulhos	R\$ 157,802.1	1139	-10.72
sao bernardo do campo	R\$ 116,155.1	898	-10.57
santos	R\$ 108,809.8	685	-9.80
santo andre	R\$ 101,090.4	765	-11.71
osasco	R\$ 90,386.1	712	-11.17
jundiai	R\$ 88,967.7	540	-10.55
sao jose dos campos	R\$ 86,148.8	658	-10.88
sorocaba	R\$ 82,525.0	601	-11.11
ribeirao preto	R\$ 71,841.4	486	-13.18
piracicaba	R\$ 57,818.1	358	-12.42
barueri	R\$ 56,713.7	414	-10.82







Recommendations			
Sales & Revenue	Logistics & Delivery	Customer Satisfaction	Seller
Double down on marketing and seller acquisition in SP to solidify market leadership, while investigating the root cause of the September sales drop to implement targeted promotional campaigns next year.	Investigate logistics partners and fulfillment centers servicing AP to reduce severe delivery bottlenecks, while preparing operational capacity for high-delay risk periods in March and November.	Address the subset of 1-star reviews (11K) by targeting the primary driver of dissatisfaction—delivery delays. Improving fulfillment speed for lagging orders will directly boost overall customer retention and score.	Expand seller onboarding outside SP to reduce geographic risk, and provide logistics support to low-performing sellers to protect customer ratings.
	Category	Product	Payment
	Focus promotions on top drivers (health beauty & bed bath table), while optimizing packaging and shipping for high-freight categories like home comfort.	Analyze high Freight % SKUs to optimize packaging or shipping terms	Maintain strong integrations and incentives with credit card partners (such as cashback or zero-interest installment promos)